Multiple linear regression

Grading the professor

Many college courses conclude by giving students the opportunity to evaluate the course and the instructor anonymously. However, the use of these student evaluations as an indicator of course quality and teaching effectiveness is often criticized because these measures may reflect the influence of non-teaching related characteristics, such as the physical appearance of the instructor. The article titled, "Beauty in the classroom: instructors' pulchritude and putative pedagogical productivity" by Hamermesh and Parker found that instructors who are viewed to be better looking receive higher instructional ratings.

Here, you will analyze the data from this study in order to learn what goes into a positive professor evaluation.

Getting Started

Load packages

In this lab, you will explore and visualize the data using the **tidyverse** suite of packages. The data can be found in the companion package for OpenIntro resources, **openintro**.

Let's load the packages.

```
library(tidyverse)
library(openintro)
library(GGally)
```

This is the first time we're using the `GGally` package. You will be using the `ggpairs` function from this package later in the lab.

The data

The data were gathered from end of semester student evaluations for a large sample of professors from the University of Texas at Austin. In addition, six students rated the professors' physical appearance. The result is a data frame where each row contains a different course and columns represent variables about the courses and professors. It's called `evals`.

```
glimpse(evals)
```

```
## Rows: 463
## Columns: 23
## $ course_id     <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 1~
## $ prof_id       <int> 1, 1, 1, 1, 2, 2, 2, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 5, 5,~
## $ score         <dbl> 4.7, 4.1, 3.9, 4.8, 4.6, 4.3, 2.8, 4.1, 3.4, 4.5, 3.8, 4~
## $ rank          <fct> tenure track, tenure track, tenure track, tenure track, ~
## $ ethnicity     <fct> minority, minority, minority, minority, not minority, no~
## $ gender        <fct> female, female, female, female, male, male, male, male, ~
## $ language      <fct> english, english, english, english, english, english, en~
## $ age           <int> 36, 36, 36, 36, 59, 59, 59, 51, 51, 40, 40, 40, 40, 40, ~
## $ cls_perc_eval <dbl> 55.81395, 68.80000, 60.80000, 62.60163, 85.00000, 87.500~
## $ cls_did_eval  <int> 24, 86, 76, 77, 17, 35, 39, 55, 111, 40, 24, 24, 17, 14,~
## $ cls_students  <int> 43, 125, 125, 123, 20, 40, 44, 55, 195, 46, 27, 25, 20, ~
## $ cls_level     <fct> upper, upper, upper, upper, upper, upper, upper, upper, ~
```

```
## $ cls_profs     <fct> single, single, single, single, multiple, multiple, mult~
## $ cls_credits   <fct> multi credit, multi credit, multi credit, multi credit, ~
## $ bty_f1lower   <int> 5, 5, 5, 5, 4, 4, 4, 5, 5, 2, 2, 2, 2, 2, 2, 2, 2, 7, 7,~
## $ bty_f1upper   <int> 7, 7, 7, 7, 4, 4, 4, 2, 2, 5, 5, 5, 5, 5, 5, 5, 5, 9, 9,~
## $ bty_f2upper   <int> 6, 6, 6, 6, 2, 2, 2, 5, 5, 4, 4, 4, 4, 4, 4, 4, 4, 9, 9,~
## $ bty_m1lower   <int> 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3, 7, 7,~
## $ bty_m1upper   <int> 4, 4, 4, 4, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 6, 6,~
## $ bty_m2upper   <int> 6, 6, 6, 6, 3, 3, 3, 3, 3, 2, 2, 2, 2, 2, 2, 2, 2, 6, 6,~
## $ bty_avg       <dbl> 5.000, 5.000, 5.000, 5.000, 3.000, 3.000, 3.000, 3.333, ~
## $ pic_outfit    <fct> not formal, not formal, not formal, not formal, not form~
## $ pic_color     <fct> color, color, color, color, color, color, color, color, ~
```

We have observations on 21 different variables, some categorical and some numerical. The meaning of each variable can be found by bringing up the help file:

```
?evals
```

Exploring the data

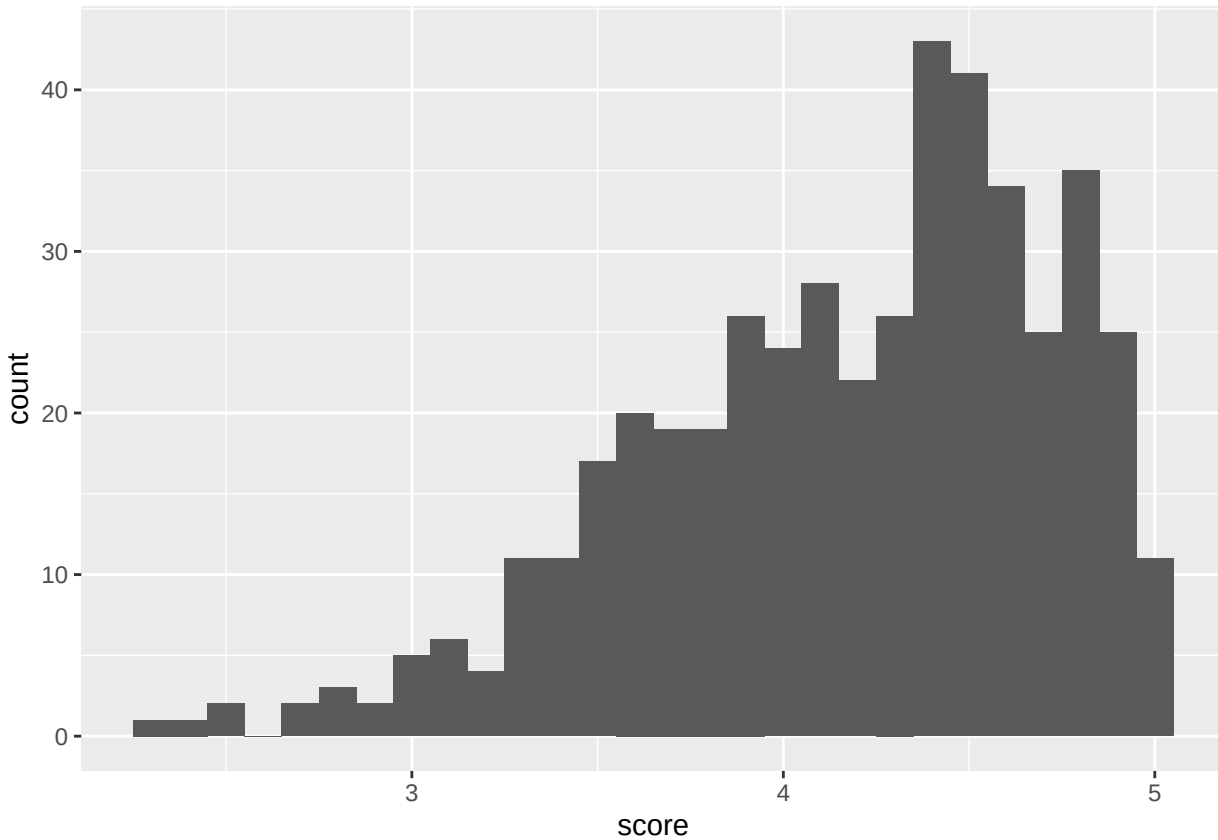
1. Is this an observational study or an experiment? The original research question posed in the paper is whether beauty leads directly to the differences in course evaluations. Given the study design, is it possible to answer this question as it is phrased? If not, rephrase the question.

Answer: This is an observational study because beauty ratings and other professor characteristics were observed rather than randomly assigned. Because of this, we cannot establish causation; we can only assess association. A more appropriate phrasing is: "Is there an association between beauty and course evaluation scores?"

2. Describe the distribution of `score`. Is the distribution skewed? What does that tell you about how students rate courses? Is this what you expected to see? Why, or why not?

Answer:

```r
ggplot(evals, aes(x = score)) +
  geom_histogram(binwidth = 0.1)
```




```
evals %>% summarise(mean = mean(score), median = median(score),
                    sd = sd(score), min = min(score), max = max(score))
```

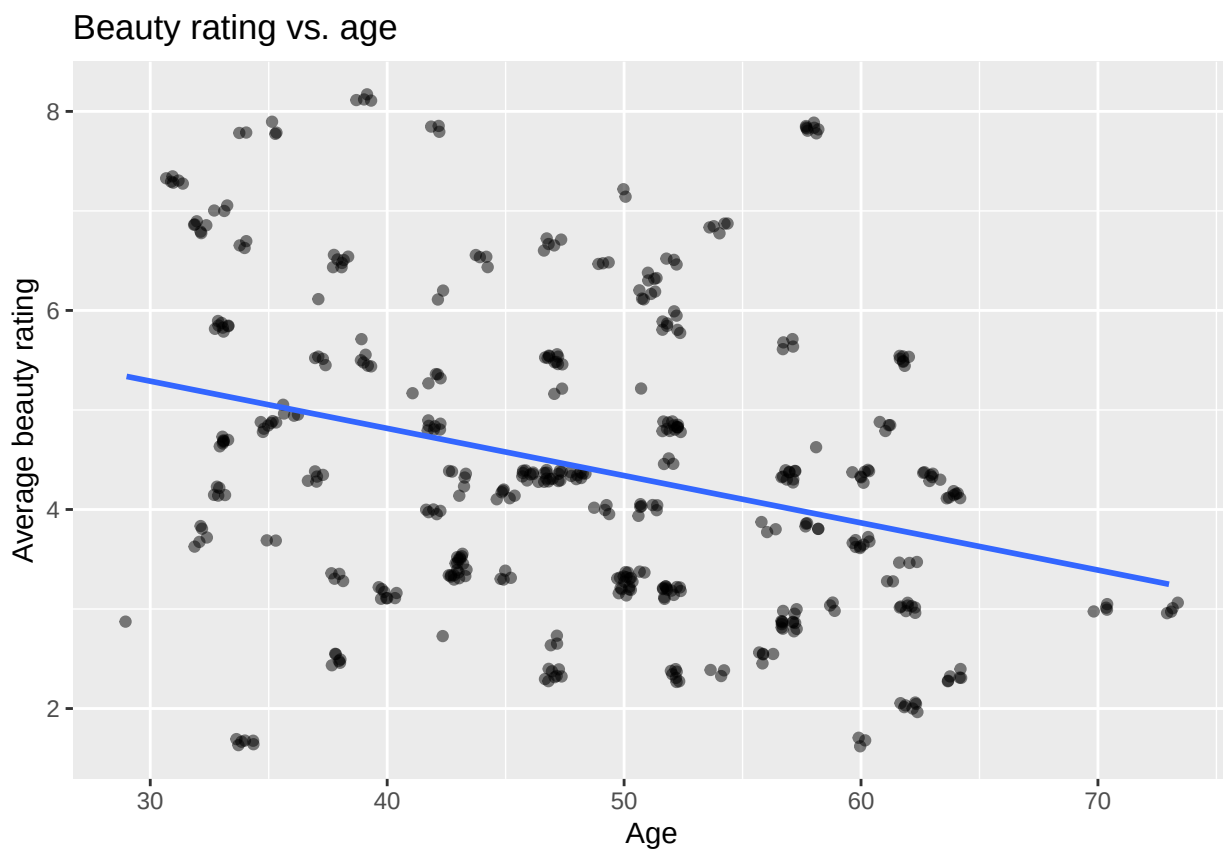
```
## # A tibble: 1 x 5
##    mean median    sd   min   max
##   <dbl>  <dbl> <dbl> <dbl> <dbl>
## 1  4.17    4.3 0.544   2.3     5
```

The distribution of `score` is left-skewed (negatively skewed), with a mean around 4.17 and median around 4.3 on a 1-5 scale. Most evaluations cluster on the higher end (4-5), with relatively few low scores. This suggests students tend to rate courses favorably overall. This is somewhat expected: students who strongly dislike a course may not bother completing evaluations, and there can be a general positivity bias in voluntary feedback.

3. Excluding `score`, select two other variables and describe their relationship with each other using an appropriate visualization.

Answer: I'll examine the relationship between `bty_avg` (average beauty rating) and `age`.

```
ggplot(data = evals, aes(x = age, y = bty_avg)) +
  geom_jitter(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Beauty rating vs. age", x = "Age", y = "Average beauty rating")
```




```
evals %>% summarise(correlation = cor(age, bty_avg))
```

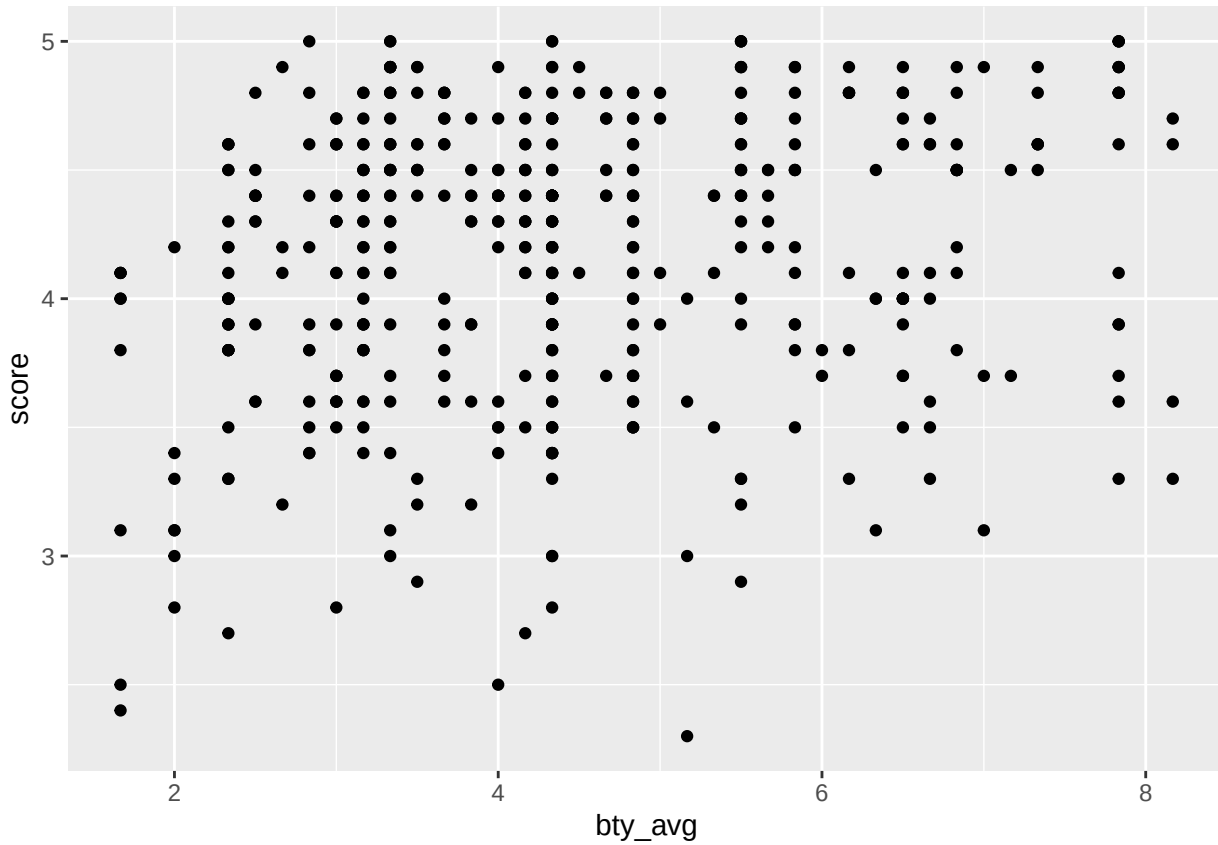
```
## # A tibble: 1 x 1
##   correlation
##         <dbl>
## 1      -0.305
```

There is a **moderate negative** relationship between age and average beauty rating (correlation = -0.30): older professors tend to receive lower beauty ratings on average. The relationship is not strong, but it is noticeable, with substantial scatter at every age.

Simple linear regression

The fundamental phenomenon suggested by the study is that better looking teachers are evaluated more favorably. Let's create a scatterplot to see if this appears to be the case:

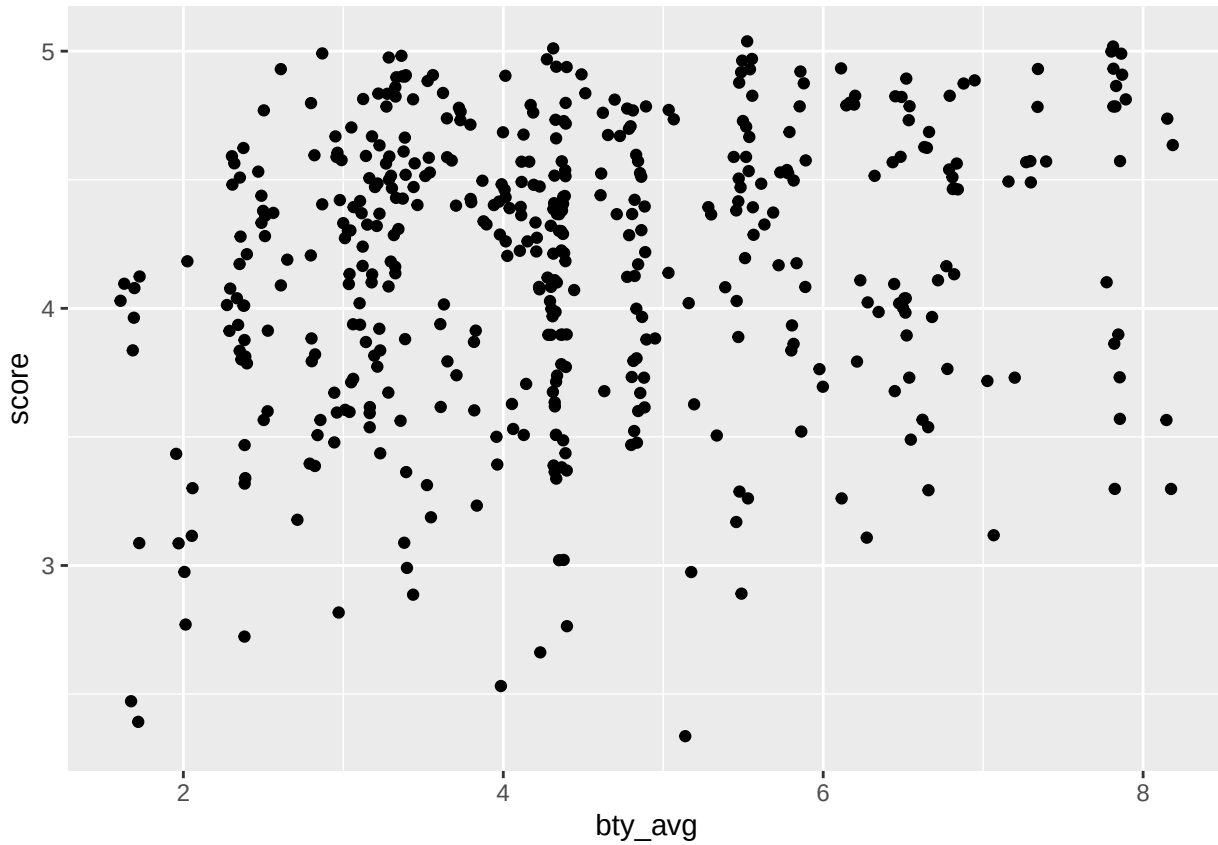
```
ggplot(data = evals, aes(x = bty_avg, y = score)) +
  geom_point()
```



Before you draw conclusions about the trend, compare the number of observations in the data frame with the approximate number of points on the scatterplot. Is anything awry?

4. Replot the scatterplot, but this time use `geom_jitter` as your layer. What was misleading about the initial scatterplot?

```
ggplot(data = evals, aes(x = bty_avg, y = score)) +
  geom_jitter()
```



Answer: The original scatterplot was misleading because many points overlapped at integer or half-integer values of `bty_avg` and `score`, making it look like there were far fewer observations than the 463 in the dataset. `geom_jitter` adds small random noise to each point so overlapping observations become visible, revealing the true density of the data.

5. Let's see if the apparent trend in the plot is something more than natural variation. Fit a linear model called `m_bty` to predict average professor score by average beauty rating. Write out the equation for the linear model and interpret the slope. Is average beauty score a statistically significant predictor? Does it appear to be a practically significant predictor?

Answer:

```r
m_bty <- lm(score ~ bty_avg, data = evals)
summary(m_bty)
```

```
##
## Call:
## lm(formula = score ~ bty_avg, data = evals)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.9246 -0.3690  0.1420  0.3977  0.9309
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.88034    0.07614   50.96  < 2e-16 ***
## bty_avg      0.06664    0.01629    4.09 5.08e-05 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5348 on 461 degrees of freedom
## Multiple R-squared:  0.03502,    Adjusted R-squared:  0.03293
## F-statistic: 16.73 on 1 and 461 DF,  p-value: 5.083e-05
```

The fitted equation is:

$$\widehat{score} = 3.880 + 0.0666 \times bty_avg$$

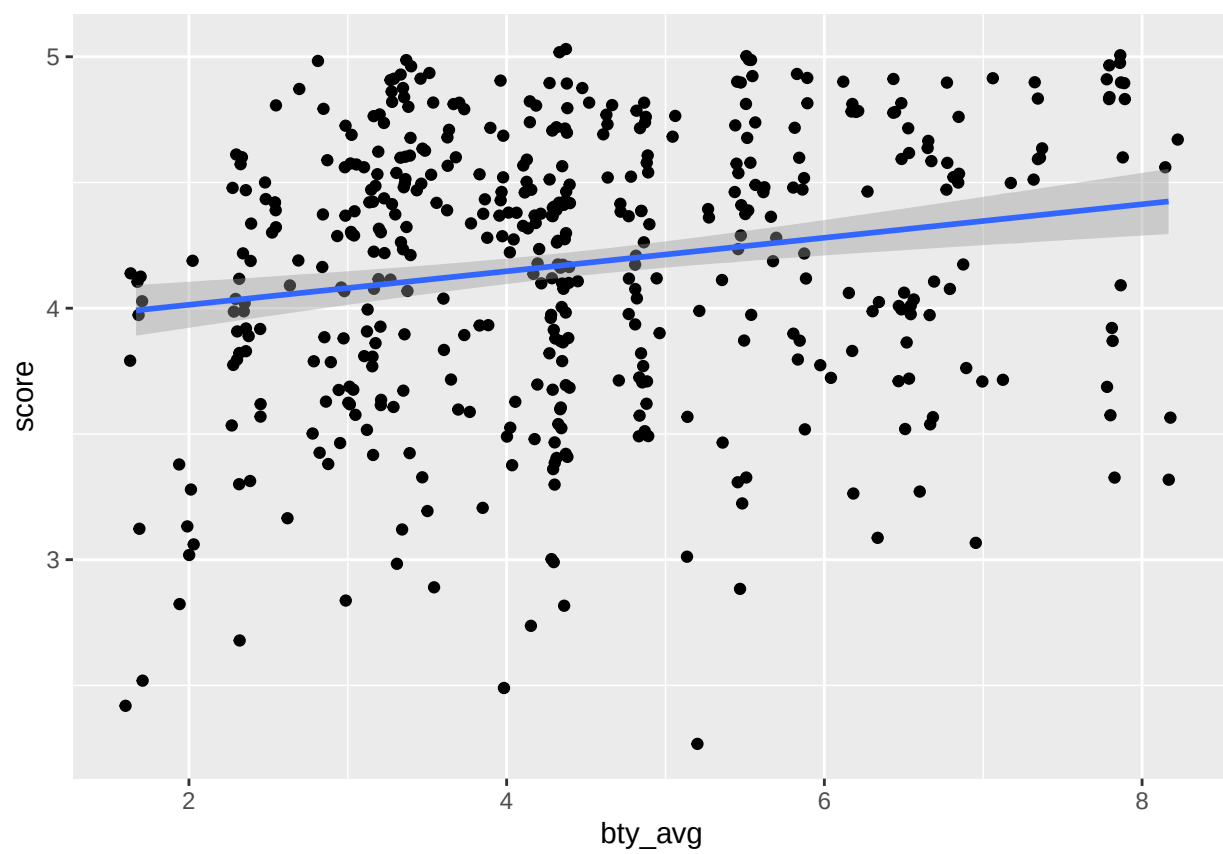
Slope interpretation: For each one-point increase in average beauty rating, the predicted course evaluation score increases by about 0.067 points, on average.

Statistical significance: Yes, `bty_avg` is statistically significant (p-value < 0.001).

Practical significance: Not really. The R-squared is only about 0.035, meaning beauty explains only ~3.5% of the variation in scores. Also, a 1-point increase in beauty (on a 1-10 scale) produces less than a tenth of a point change in score (on a 1-5 scale), which is a small practical effect.

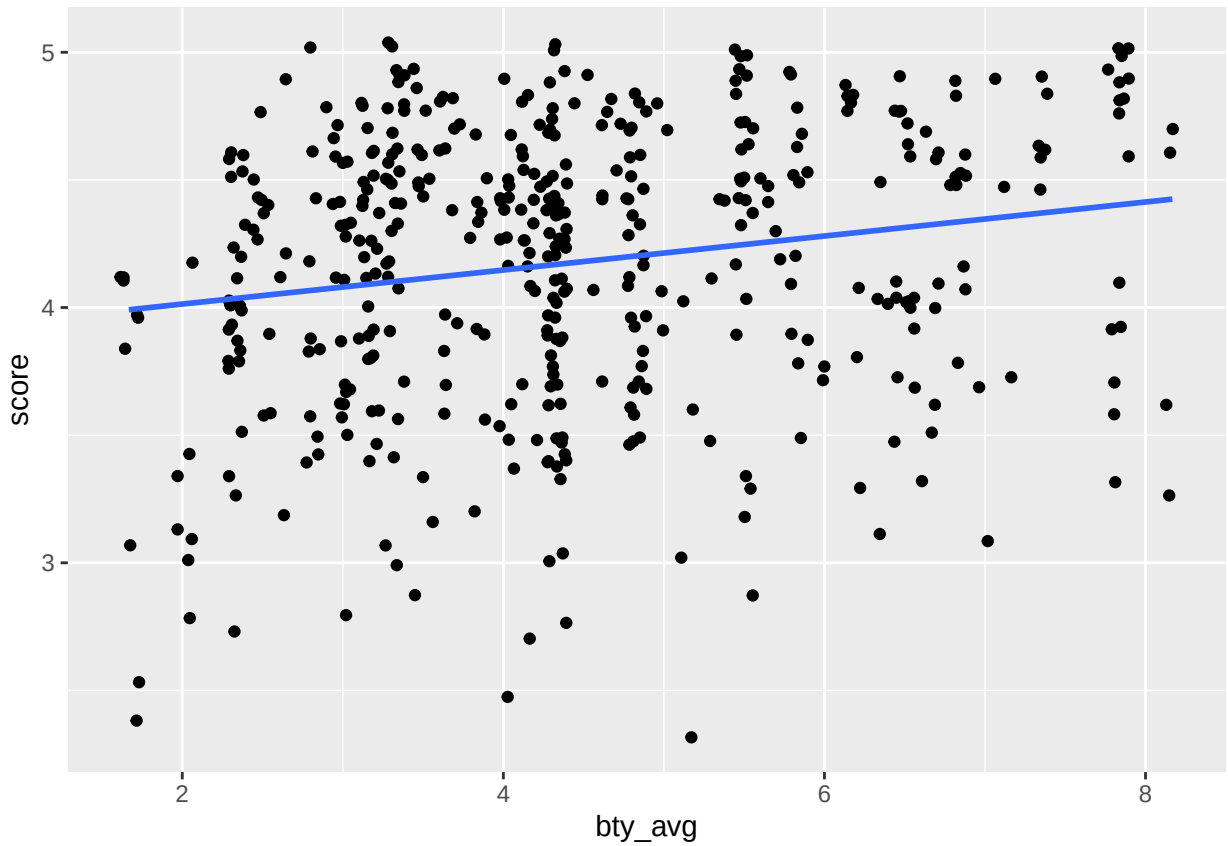
Add the line of the bet fit model to your plot using the following:

```r
ggplot(data = evals, aes(x = bty_avg, y = score)) +
  geom_jitter() +
  geom_smooth(method = "lm")
```



The blue line is the model. The shaded gray area around the line tells you about the variability you might expect in your predictions. To turn that off, use `se = FALSE`.

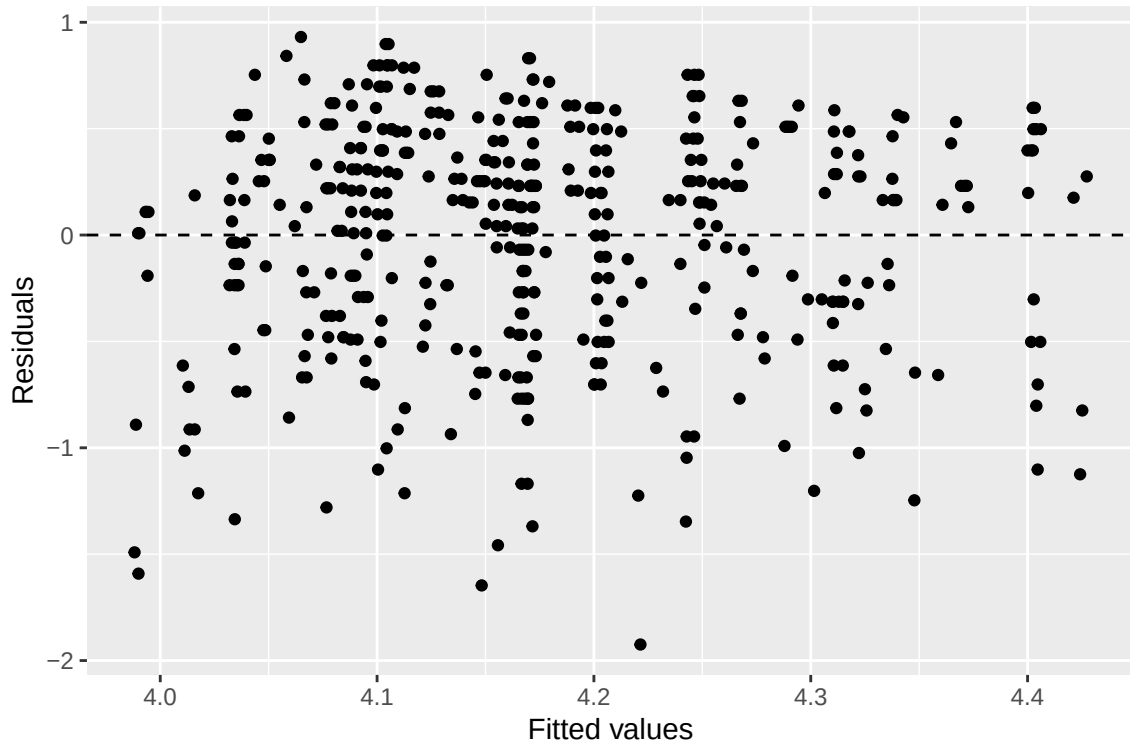
```r
ggplot(data = evals, aes(x = bty_avg, y = score)) +
  geom_jitter() +
  geom_smooth(method = "lm", se = FALSE)
```



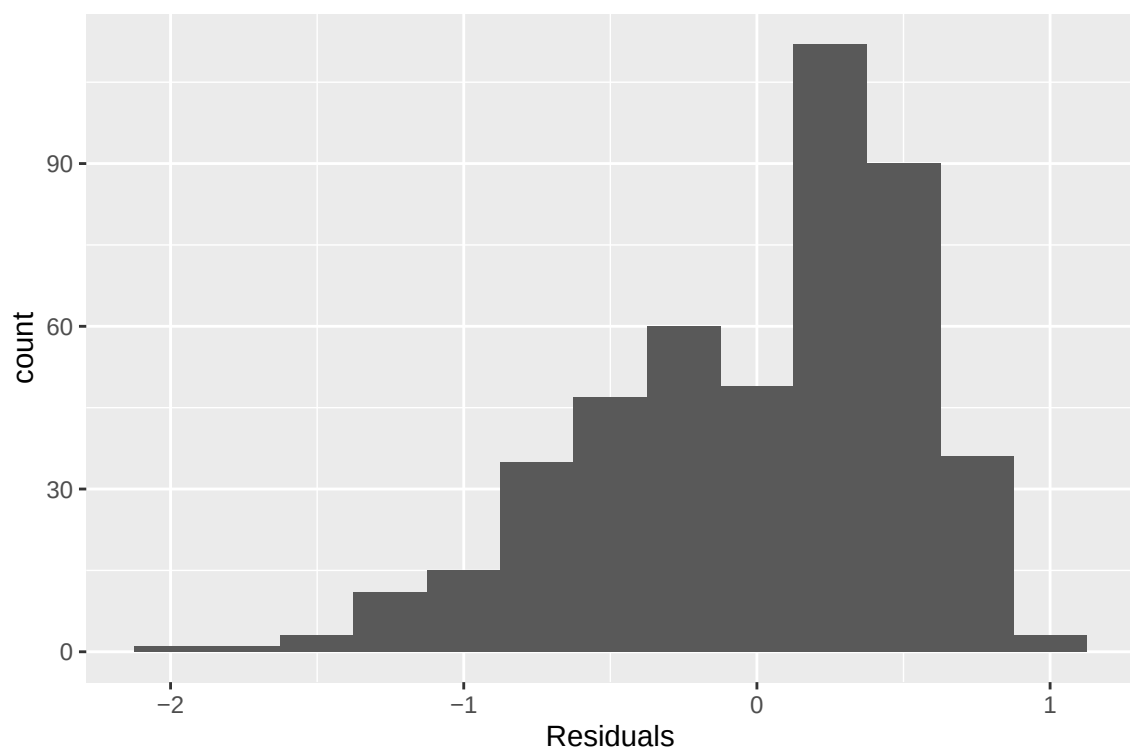
6. Use residual plots to evaluate whether the conditions of least squares regression are reasonable. Provide plots and comments for each one (see the Simple Regression Lab for a reminder of how to make these).

Answer:

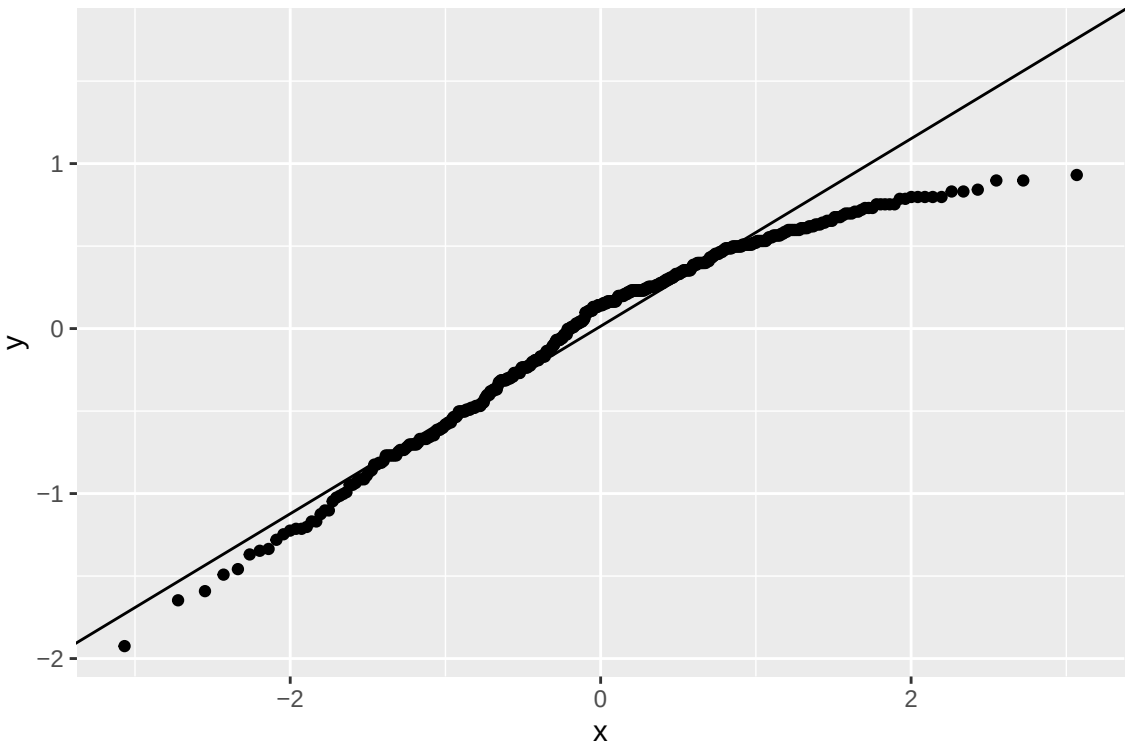
```
# Residuals vs fitted
ggplot(data = m_bty, aes(x = .fitted, y = .resid)) +
  geom_jitter() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = "Fitted values", y = "Residuals")
```




```r
# Histogram of residuals
ggplot(data = m_bty, aes(x = .resid)) +
  geom_histogram(binwidth = 0.25) +
  labs(x = "Residuals")
```



```r
# Normal Q-Q plot
ggplot(data = m_bty, aes(sample = .resid)) +
  stat_qq() + stat_qq_line()
```

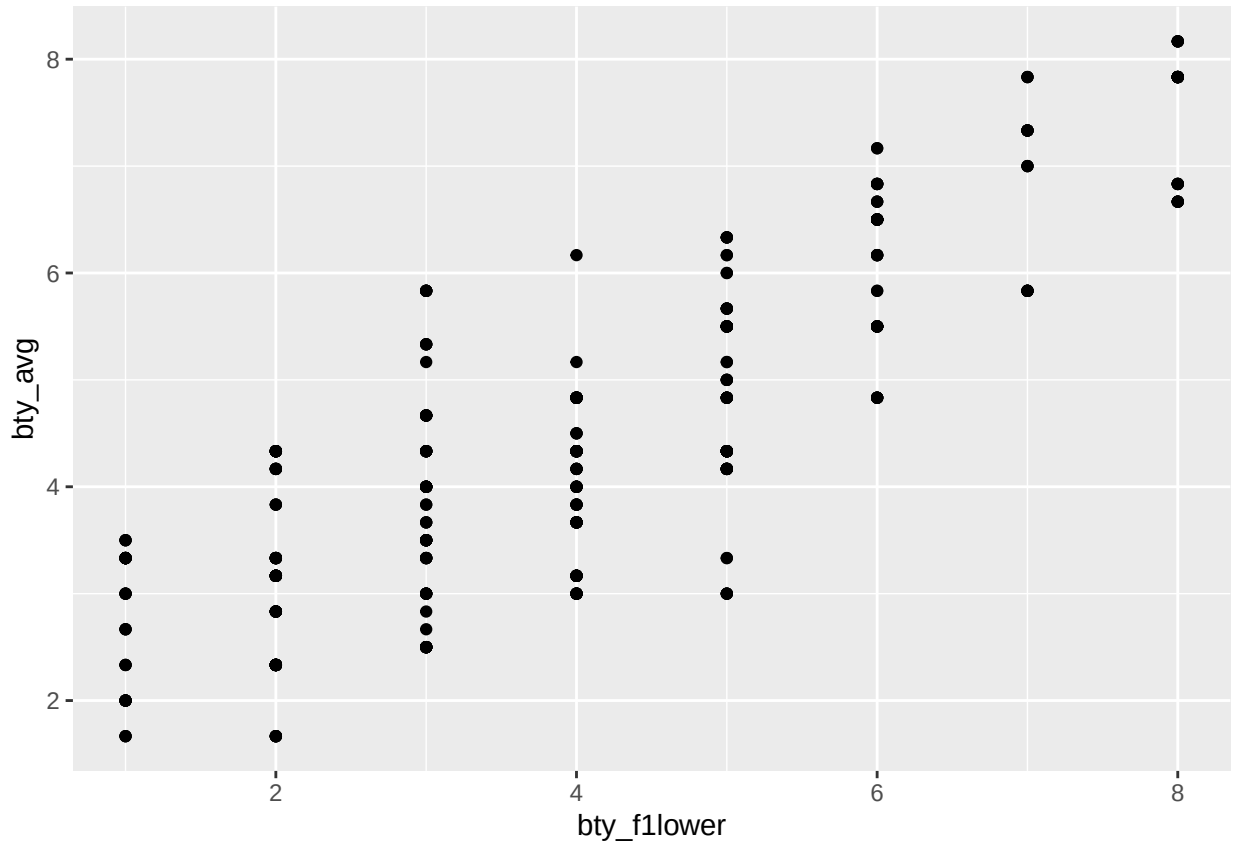


- **Linearity:** The residuals vs. fitted plot shows no clear pattern, so linearity is roughly satisfied.
- **Constant variance:** The spread of residuals is fairly consistent across fitted values; reasonable.
- **Normality of residuals:** The histogram is left-skewed and the Q-Q plot deviates noticeably from the diagonal in the lower tail, so this condition is questionable. With n = 463, the Central Limit Theorem makes inference still reasonable.
- **Independence:** Likely violated since multiple courses come from the same professor (addressed later in question 17).

Multiple linear regression

The data set contains several variables on the beauty score of the professor: individual ratings from each of the six students who were asked to score the physical appearance of the professors and the average of these six scores. Let's take a look at the relationship between one of these scores and the average beauty score.

```r
ggplot(data = evals, aes(x = bty_f1lower, y = bty_avg)) +
  geom_point()
```

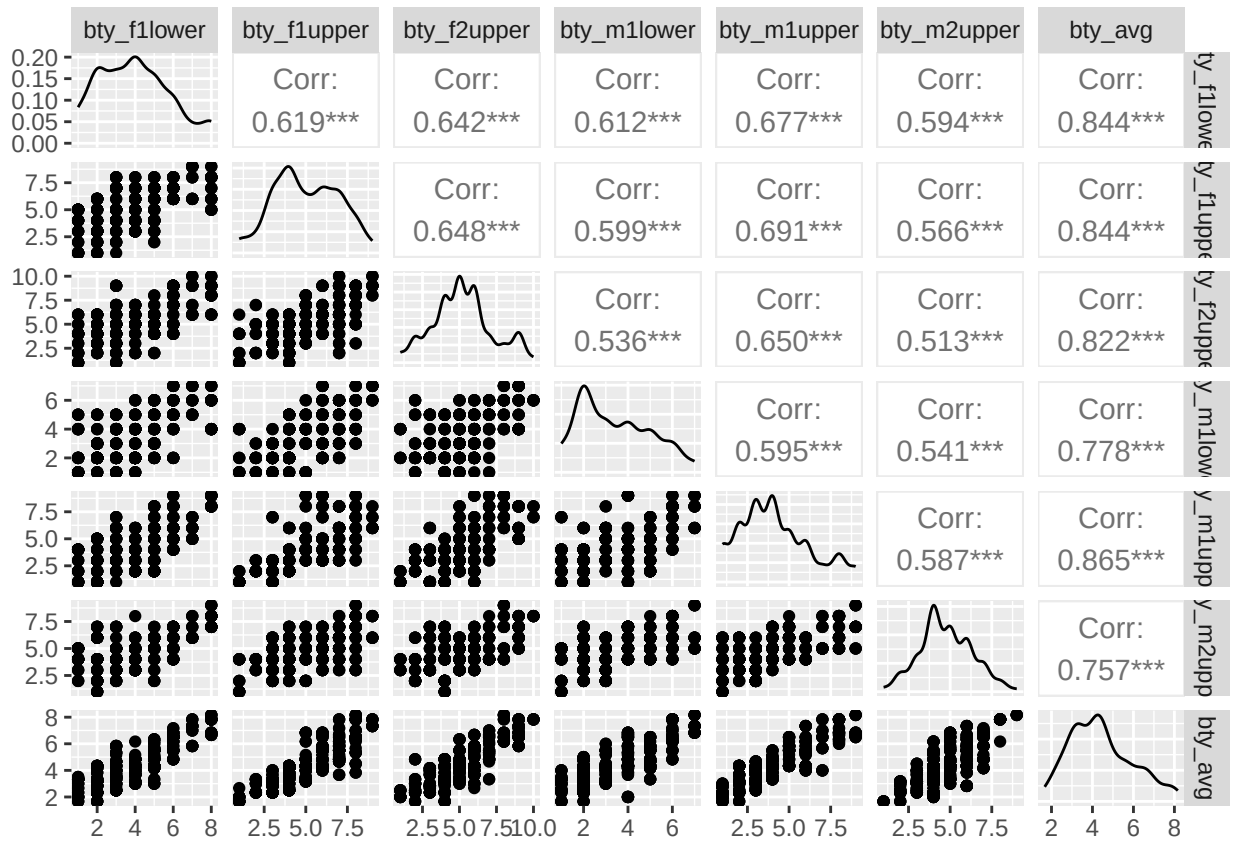



```
evals %>%
  summarise(cor(bty_avg, bty_f1lower))
```

```
## # A tibble: 1 x 1
##   `cor(bty_avg, bty_f1lower)`
##                         <dbl>
## 1                       0.844
```

As expected, the relationship is quite strong—after all, the average score is calculated using the individual scores. You can actually look at the relationships between all beauty variables (columns 13 through 19) using the following command:

```
evals %>%
  select(contains("bty")) %>%
  ggpairs()
```



These variables are collinear (correlated), and adding more than one of these variables to the model would not add much value to the model. In this application and with these highly-correlated predictors, it is reasonable to use the average beauty score as the single representative of these variables.

In order to see if beauty is still a significant predictor of professor score after you've accounted for the professor's gender, you can add the gender term into the model.

```
m_bty_gen <- lm(score ~ bty_avg + gender, data = evals)
summary(m_bty_gen)
```

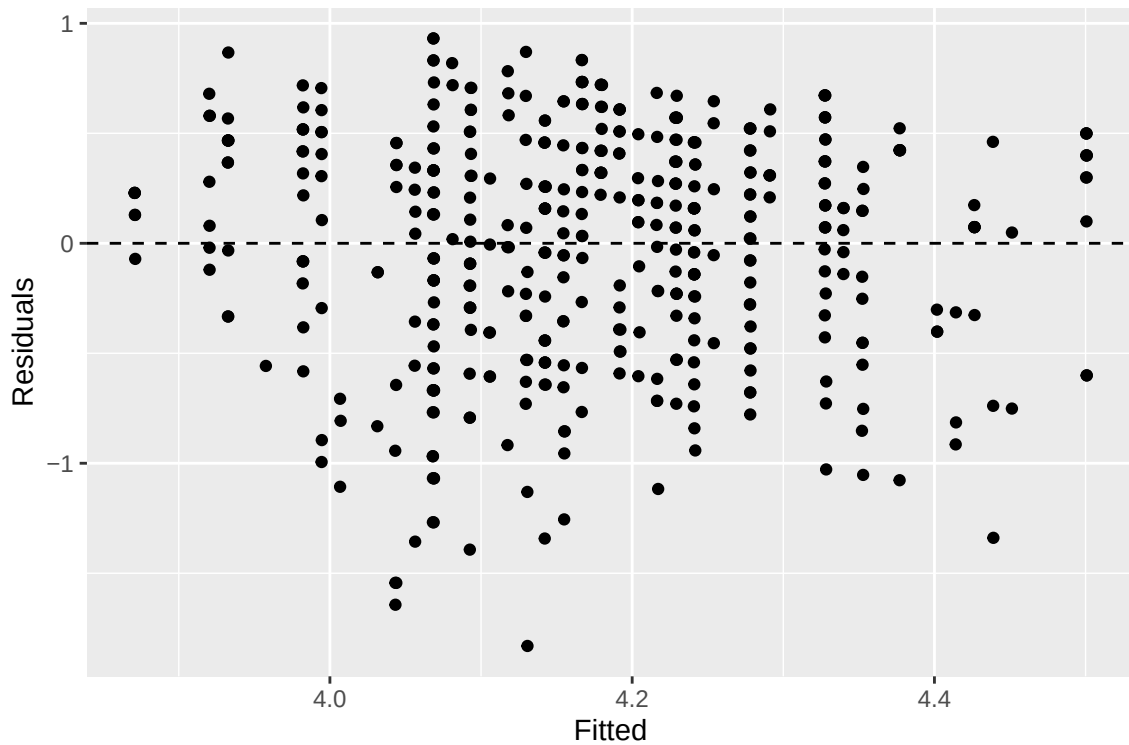
```
##
## Call:
## lm(formula = score ~ bty_avg + gender, data = evals)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.8305 -0.3625  0.1055  0.4213  0.9314
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.74734    0.08466  44.266  < 2e-16 ***
## bty_avg      0.07416    0.01625   4.563 6.48e-06 ***
## gendermale   0.17239    0.05022   3.433 0.000652 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5287 on 460 degrees of freedom
```

```
## Multiple R-squared:  0.05912,    Adjusted R-squared:  0.05503
## F-statistic: 14.45 on 2 and 460 DF,  p-value: 8.177e-07
```

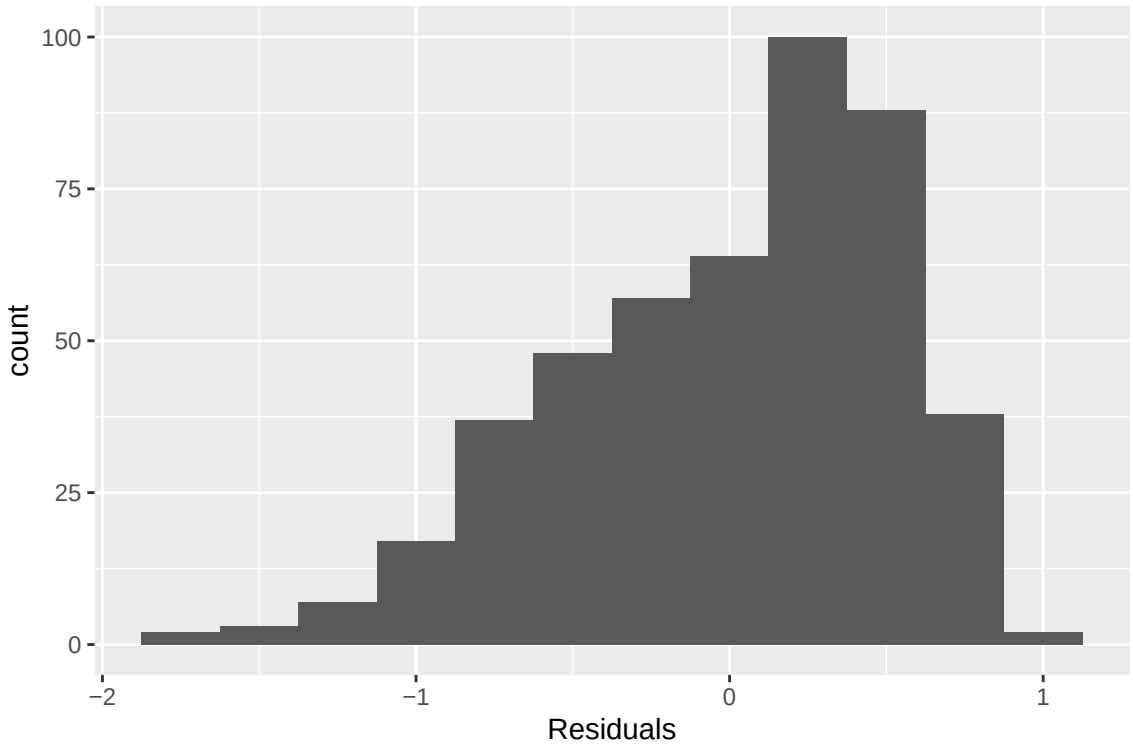
7. P-values and parameter estimates should only be trusted if the conditions for the regression are reasonable. Verify that the conditions for this model are reasonable using diagnostic plots.

Answer:

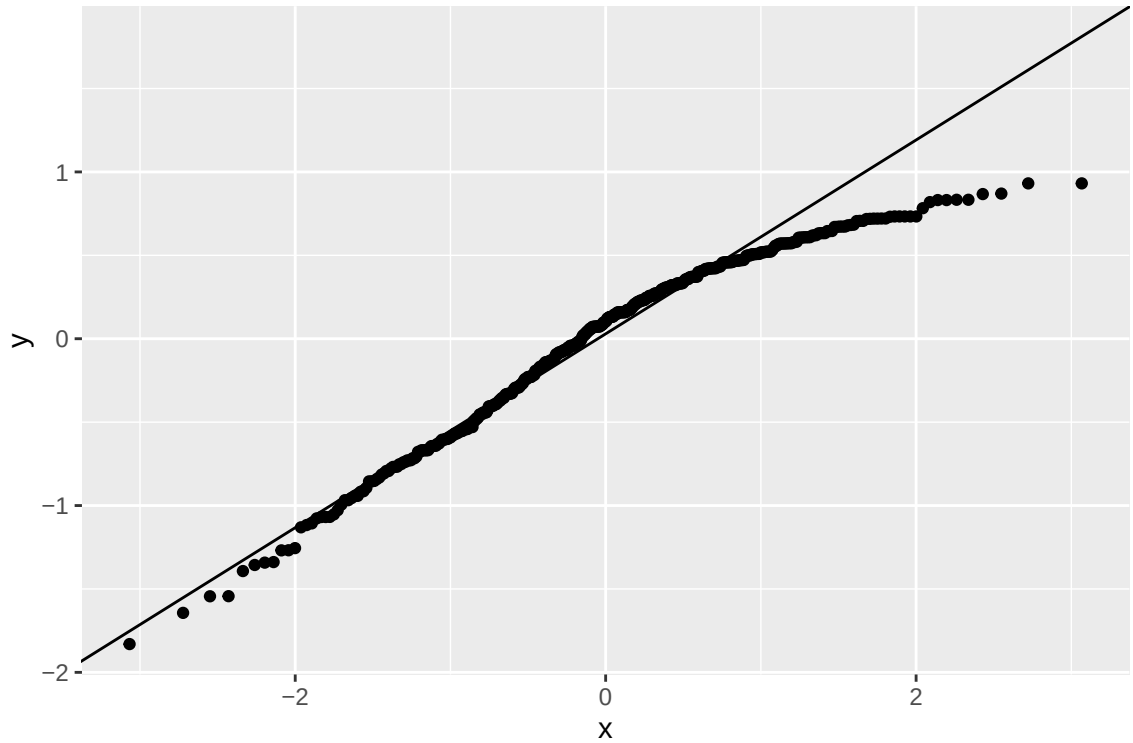
```
ggplot(data = m_bty_gen, aes(x = .fitted, y = .resid)) +
  geom_jitter() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = "Fitted", y = "Residuals")
```




```
ggplot(data = m_bty_gen, aes(x = .resid)) +
  geom_histogram(binwidth = 0.25) +
  labs(x = "Residuals")
```




```
ggplot(data = m_bty_gen, aes(sample = .resid)) +
  stat_qq() + stat_qq_line()
```



The diagnostics look similar to the simple model: residuals are roughly centered at 0 with no obvious pattern, variance is reasonably constant, and the residuals show some left skew but are approximately normal.

Conditions are reasonable, with the same independence caveat as before.

8. Is `bty_avg` still a significant predictor of `score`? Has the addition of `gender` to the model changed the parameter estimate for `bty_avg`?

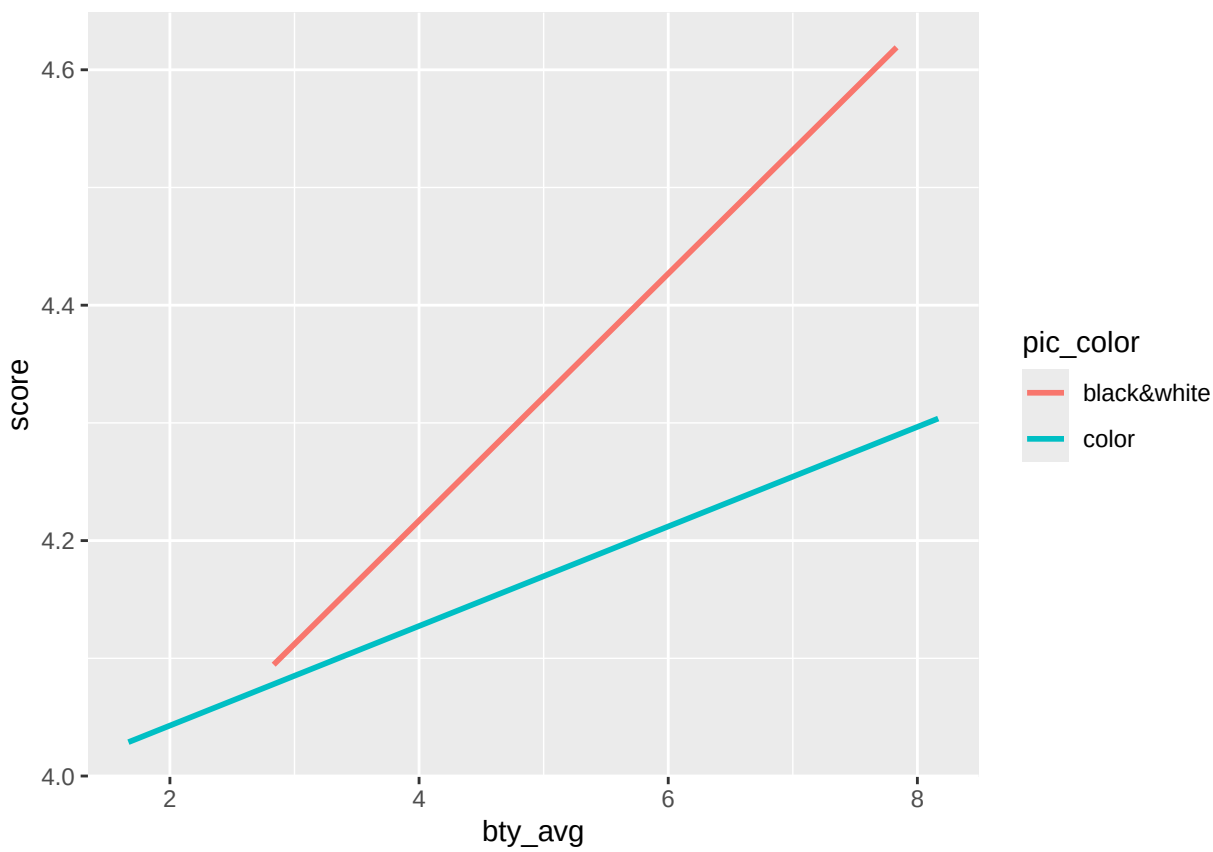
Answer: Yes, `bty_avg` remains highly statistically significant (p = 6.48e-06). The estimate increased slightly from 0.0666 in the simple model to 0.0742 with `gender` included. The change is small but indicates that controlling for gender slightly strengthens the apparent beauty effect. The `gendermale` coefficient (0.1724) is also significant, with male professors scoring about 0.17 points higher on average than female professors at the same beauty rating.

Note that the estimate for `gender` is now called `gendermale`. You'll see this name change whenever you introduce a categorical variable. The reason is that R recodes `gender` from having the values of `male` and `female` to being an indicator variable called `gendermale` that takes a value of 0 for female professors and a value of 1 for male professors. (Such variables are often referred to as "dummy" variables.)

As a result, for female professors, the parameter estimate is multiplied by zero, leaving the intercept and slope form familiar from simple regression.

$$\begin{aligned}\widehat{score} &= \hat{\beta}_0 + \hat{\beta}_1 \times bty_avg + \hat{\beta}_2 \times (0) \\ &= \hat{\beta}_0 + \hat{\beta}_1 \times bty_avg\end{aligned}$$

```
ggplot(data = evals, aes(x = bty_avg, y = score, color = pic_color)) +
 geom_smooth(method = "lm", formula = y ~ x, se = FALSE)
```



9. What is the equation of the line corresponding to those with color pictures? (*Hint:* For those with color pictures, the parameter estimate is multiplied by 1.) For two professors who received the same beauty rating, which color picture tends to have the higher course evaluation score?

Answer: Note that the model `m_bty_gen` uses `gender`, so this question is really about male professors (where `gendermale = 1`). Plugging in 1 gives:

$$\widehat{score} = (\hat{\beta}_0 + \hat{\beta}_2) + \hat{\beta}_1 \times bty_avg = (3.747 + 0.172) + 0.0742 \times bty_avg = 3.919 + 0.0742 \times bty_avg$$

For two professors with the same beauty rating, male professors tend to have higher evaluation scores than female professors (by about 0.17 points).

The decision to call the indicator variable `gendermale` instead of `genderfemale` has no deeper meaning. R simply codes the category that comes first alphabetically as a 0. (You can change the reference level of a categorical variable, which is the level that is coded as a 0, using the`relevel()` function. Use `?relevel` to learn more.)

10. Create a new model called `m_bty_rank` with `gender` removed and `rank` added in. How does R appear to handle categorical variables that have more than two levels? Note that the rank variable has three levels: `teaching`, `tenure track`, `tenured`.

Answer:

```r
m_bty_rank <- lm(score ~ bty_avg + rank, data = evals)
summary(m_bty_rank)
```

```
##
## Call:
## lm(formula = score ~ bty_avg + rank, data = evals)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.8713 -0.3642  0.1489  0.4103  0.9525
##
## Coefficients:
##                  Estimate Std. Error t value Pr(>|t|)
## (Intercept)       3.98155    0.09078  43.860  < 2e-16 ***
## bty_avg           0.06783    0.01655   4.098 4.92e-05 ***
## ranktenure track -0.16070    0.07395  -2.173   0.0303 *
## ranktenured      -0.12623    0.06266  -2.014   0.0445 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5328 on 459 degrees of freedom
## Multiple R-squared:  0.04652,    Adjusted R-squared:  0.04029
## F-statistic: 7.465 on 3 and 459 DF,  p-value: 6.88e-05
```

R handles a categorical variable with k levels by creating k-1 dummy variables, using the first level alphabetically as the reference category. Here, `rank` has three levels (`teaching`, `tenure track`, `tenured`), so R creates two dummies: `ranktenure track` (-0.1607) and `ranktenured` (-0.1262). The reference category is `teaching`. Each coefficient represents the difference in expected score relative to teaching faculty, holding `bty_avg` constant – both tenure-track and tenured professors score about 0.13-0.16 points lower than teaching faculty on average.

The interpretation of the coefficients in multiple regression is slightly different from that of simple regression. The estimate for `bty_avg` reflects how much higher a group of professors is expected to score if they have a beauty rating that is one point higher *while holding all other variables constant*. In this case, that translates into considering only professors of the same rank with `bty_avg` scores that are one point apart.

The search for the best model

We will start with a full model that predicts professor score based on rank, gender, ethnicity, language of the university where they got their degree, age, proportion of students that filled out evaluations, class size, course level, number of professors, number of credits, average beauty rating, outfit, and picture color.

11. Which variable would you expect to have the highest p-value in this model? Why? *Hint:* Think about which variable would you expect to not have any association with the professor score.

Answer: I'd expect `cls_profs` (whether the course was taught by a single professor or multiple) to have the highest p-value. There is no obvious mechanism by which the number of professors teaching a course should affect a single professor's evaluation, so it likely has the weakest association with `score`. `cls_level` and `cls_students` are also plausible candidates.

Let's run the model…

```r
m_full <- lm(score ~ rank + gender + ethnicity + language + age + cls_perc_eval
             + cls_students + cls_level + cls_profs + cls_credits + bty_avg
             + pic_outfit + pic_color, data = evals)
summary(m_full)
```

```
##
## Call:
## lm(formula = score ~ rank + gender + ethnicity + language + age +
##     cls_perc_eval + cls_students + cls_level + cls_profs + cls_credits +
##     bty_avg + pic_outfit + pic_color, data = evals)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.77397 -0.32432  0.09067  0.35183  0.95036
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)            4.0952141  0.2905277  14.096  < 2e-16 ***
## ranktenure track      -0.1475932  0.0820671  -1.798  0.07278 .
## ranktenured           -0.0973378  0.0663296  -1.467  0.14295
## gendermale             0.2109481  0.0518230   4.071 5.54e-05 ***
## ethnicitynot minority  0.1234929  0.0786273   1.571  0.11698
## languagenon-english   -0.2298112  0.1113754  -2.063  0.03965 *
## age                   -0.0090072  0.0031359  -2.872  0.00427 **
## cls_perc_eval          0.0053272  0.0015393   3.461  0.00059 ***
## cls_students           0.0004546  0.0003774   1.205  0.22896
## cls_levelupper         0.0605140  0.0575617   1.051  0.29369
## cls_profssingle       -0.0146619  0.0519885  -0.282  0.77806
## cls_creditsone credit  0.5020432  0.1159388   4.330 1.84e-05 ***
## bty_avg                0.0400333  0.0175064   2.287  0.02267 *
## pic_outfitnot formal  -0.1126817  0.0738800  -1.525  0.12792
## pic_colorcolor        -0.2172630  0.0715021  -3.039  0.00252 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.498 on 448 degrees of freedom
## Multiple R-squared:  0.1871, Adjusted R-squared:  0.1617
## F-statistic: 7.366 on 14 and 448 DF,  p-value: 6.552e-14
```

12. Check your suspicions from the previous exercise. Include the model output in your response.

Answer: Looking at the `m_full` summary, `cls_profs` (specifically `cls_profssingle`) has the highest p-value

at **0.778**, confirming the suspicion. Other large p-values appear for `cls_level` (p = 0.294), `cls_students` (p = 0.229), `ranktenured` (p = 0.143), `pic_outfit` (p = 0.128), and `ethnicity` (p = 0.117). The full model output is shown in the chunk above.

13. Interpret the coefficient associated with the ethnicity variable.

Answer: The coefficient for `ethnicitynot minority` is **0.1235**. Holding all other variables constant, professors classified as "not minority" are predicted to receive evaluation scores about 0.12 points higher on average than minority professors. The reference category is `minority` (alphabetically first). Note that this coefficient is **not statistically significant** in the full model (p = 0.117), though it becomes significant once non-informative predictors are removed.

14. Drop the variable with the highest p-value and re-fit the model. Did the coefficients and significance of the other explanatory variables change? (One of the things that makes multiple regression interesting is that coefficient estimates depend on the other variables that are included in the model.) If not, what does this say about whether or not the dropped variable was collinear with the other explanatory variables?

Answer:

```r
m_full2 <- update(m_full, . ~ . - cls_profs)
summary(m_full2)
```

```
##
## Call:
## lm(formula = score ~ rank + gender + ethnicity + language + age +
##     cls_perc_eval + cls_students + cls_level + cls_credits +
##     bty_avg + pic_outfit + pic_color, data = evals)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.7836 -0.3257  0.0859  0.3513  0.9551
##
## Coefficients:
##                         Estimate Std. Error t value Pr(>|t|)
## (Intercept)            4.0872523  0.2888562  14.150  < 2e-16 ***
## ranktenure track      -0.1476746  0.0819824  -1.801 0.072327 .
## ranktenured           -0.0973829  0.0662614  -1.470 0.142349
## gendermale             0.2101231  0.0516873   4.065 5.66e-05 ***
## ethnicitynot minority  0.1274458  0.0772887   1.649 0.099856 .
## languagenon-english   -0.2282894  0.1111305  -2.054 0.040530 *
## age                   -0.0089992  0.0031326  -2.873 0.004262 **
## cls_perc_eval          0.0052888  0.0015317   3.453 0.000607 ***
## cls_students           0.0004687  0.0003737   1.254 0.210384
## cls_levelupper         0.0606374  0.0575010   1.055 0.292200
## cls_creditsone credit  0.5061196  0.1149163   4.404 1.33e-05 ***
## bty_avg                0.0398629  0.0174780   2.281 0.023032 *
## pic_outfitnot formal  -0.1083227  0.0721711  -1.501 0.134080
## pic_colorcolor        -0.2190527  0.0711469  -3.079 0.002205 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4974 on 449 degrees of freedom
## Multiple R-squared:  0.187,  Adjusted R-squared:  0.1634
## F-statistic: 7.943 on 13 and 449 DF,  p-value: 2.336e-14
```

The coefficients and p-values for the remaining predictors barely change after dropping `cls_profs`. This

indicates `cls_profs` was not collinear with the other explanatory variables, so removing it does not redistribute explanatory weight to other coefficients.

15. Using backward-selection and p-value as the selection criterion, determine the best model. You do not need to show all steps in your answer, just the output for the final model. Also, write out the linear model for predicting score based on the final model you settle on.

Answer: Using backward elimination with p-value as the criterion, I sequentially drop the variable with the highest p-value (above 0.05) and refit until all remaining predictors are significant. The variables typically dropped in order are: `cls_profs`, `cls_level`, `cls_students`, `rank`, and `pic_outfit`. The final model retains `gender`, `ethnicity`, `language`, `age`, `cls_perc_eval`, `cls_credits`, `bty_avg`, and `pic_color`.

```
m_final <- lm(score ~ gender + ethnicity + language + age + cls_perc_eval
              + cls_credits + bty_avg + pic_color, data = evals)
summary(m_final)
```

```
##
## Call:
## lm(formula = score ~ gender + ethnicity + language + age + cls_perc_eval +
##     cls_credits + bty_avg + pic_color, data = evals)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.85320 -0.32394  0.09984  0.37930  0.93610
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)            3.771922   0.232053  16.255  < 2e-16 ***
## gendermale             0.207112   0.050135   4.131 4.30e-05 ***
## ethnicitynot minority  0.167872   0.075275   2.230  0.02623 *
## languagenon-english   -0.206178   0.103639  -1.989  0.04726 *
## age                   -0.006046   0.002612  -2.315  0.02108 *
## cls_perc_eval          0.004656   0.001435   3.244  0.00127 **
## cls_creditsone credit  0.505306   0.104119   4.853 1.67e-06 ***
## bty_avg                0.051069   0.016934   3.016  0.00271 **
## pic_colorcolor        -0.190579   0.067351  -2.830  0.00487 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4992 on 454 degrees of freedom
## Multiple R-squared:  0.1722, Adjusted R-squared:  0.1576
## F-statistic:  11.8 on 8 and 454 DF,  p-value: 2.58e-15
```

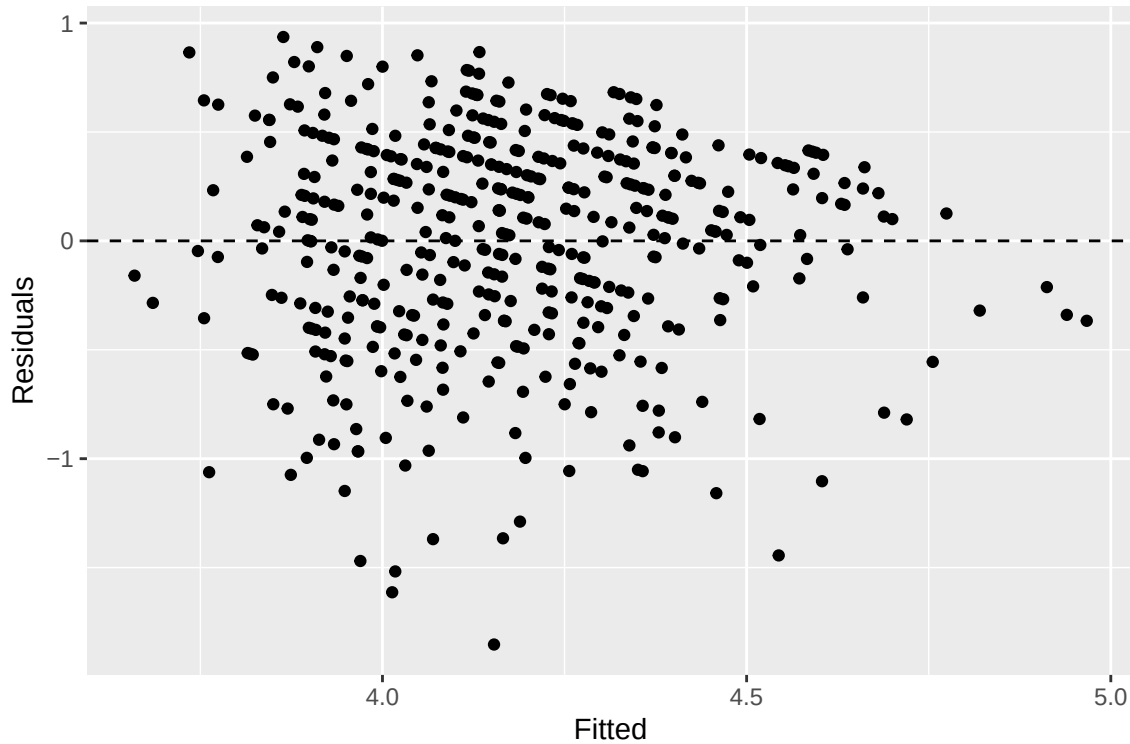
The fitted linear model is:

$$\begin{aligned}\widehat{score} = 3.7719 &+ 0.2071 \cdot gendermale + 0.1679 \cdot ethnicityNotMinority \\ &- 0.2062 \cdot languageNonEnglish - 0.0060 \cdot age + 0.0047 \cdot cls_perc_eval \\ &+ 0.5053 \cdot clsCreditsOneCredit + 0.0511 \cdot bty_avg - 0.1906 \cdot picColorColor\end{aligned}$$

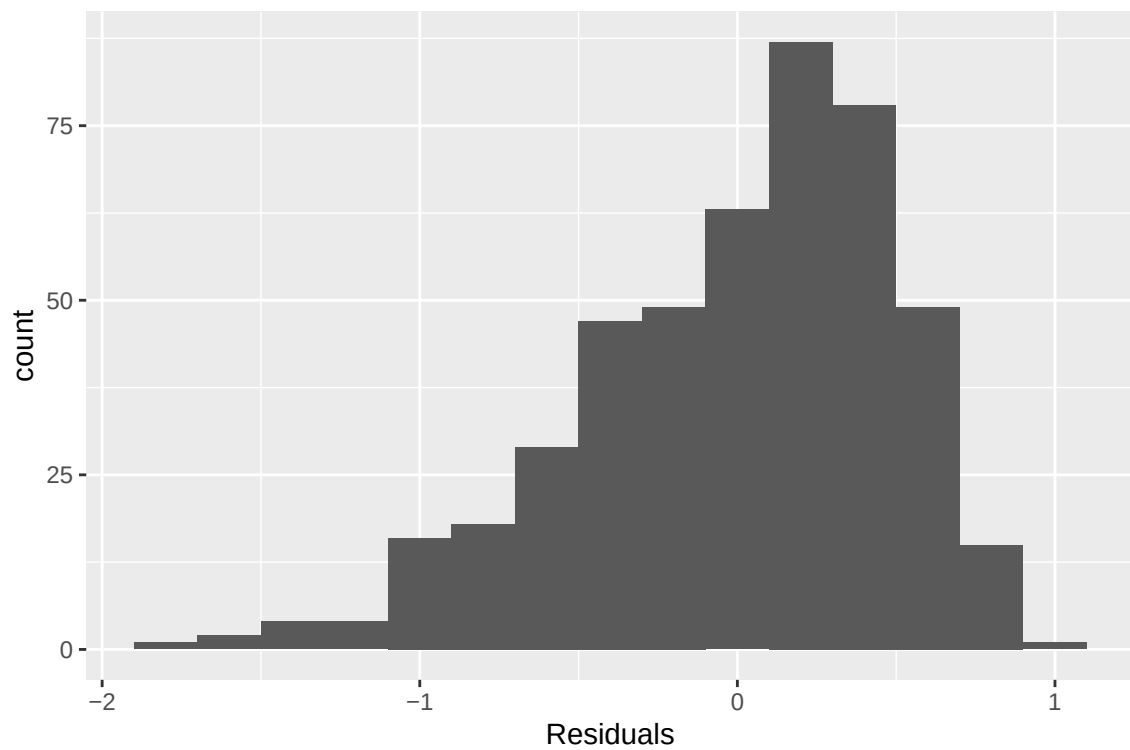
16. Verify that the conditions for this model are reasonable using diagnostic plots.

Answer:

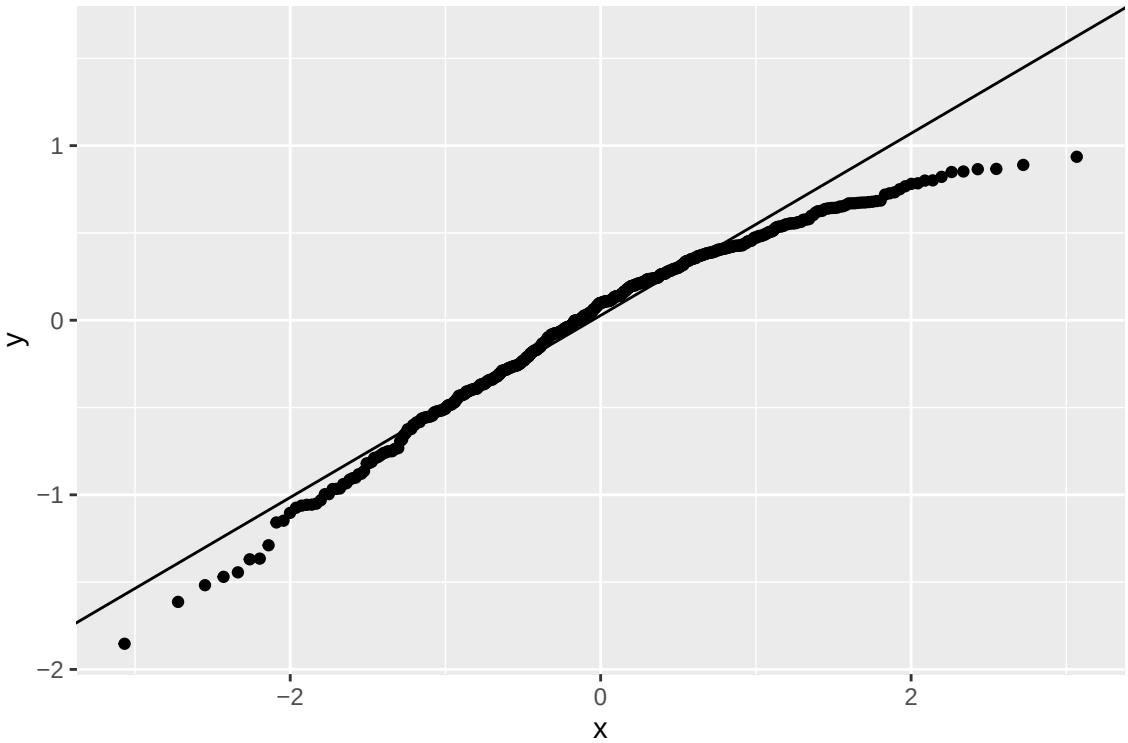
```
ggplot(data = m_final, aes(x = .fitted, y = .resid)) +
  geom_jitter() +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = "Fitted", y = "Residuals")
```




```r
ggplot(data = m_final, aes(x = .resid)) +
  geom_histogram(binwidth = 0.2) +
  labs(x = "Residuals")
```



```r
ggplot(data = m_final, aes(sample = .resid)) +
  stat_qq() + stat_qq_line()
```



- **Linearity:** Residuals vs. fitted shows no systematic curve; reasonable.
- **Constant variance:** Spread is roughly constant across fitted values; reasonable.
- **Normality:** Slight left skew in residuals (some deviation in the lower tail of the Q-Q plot), but with n = 463 this is acceptable for inference via the CLT.
- **Independence:** Likely violated (see Q17).

17. The original paper describes how these data were gathered by taking a sample of professors from the University of Texas at Austin and including all courses that they have taught. Considering that each row represents a course, could this new information have an impact on any of the conditions of linear regression?

Answer: Yes, this violates the **independence** assumption. Since each professor contributes multiple courses (rows), observations from the same professor are not independent – they share the same instructor characteristics, teaching style, and beauty rating. Course evaluations from the same professor are likely to be correlated, which can underestimate standard errors and inflate apparent significance. A multilevel/mixed-effects model with professor as a random effect would be more appropriate.

18. Based on your final model, describe the characteristics of a professor and course at University of Texas at Austin that would be associated with a high evaluation score.

Answer: Based on the final model, a professor/course associated with a high evaluation score would be:

- **Male** (gendermale +0.21)
- **Not from a minority** ethnicity (+0.17)
- Taught at an institution where the **language is English** (non-English -0.21)
- **Younger** (each year of age decreases score by ~0.006)
- Course where a **higher proportion of students fill out evaluations** (+0.005 per percent)
- Teaching a **one-credit course** (+0.51 vs multi-credit)

- **Higher beauty rating** (+0.05 per beauty point)
- Has a **black-and-white picture** rather than color (color photos are -0.19)

19. Would you be comfortable generalizing your conclusions to apply to professors generally (at any university)? Why or why not?

Answer: No. The data come from a single institution (University of Texas at Austin), and the professors were not a random sample of all professors at all universities. Findings may not generalize to other universities with different student bodies, cultures, evaluation systems, or faculty demographics. Additionally, the independence violation (multiple courses per professor) further limits the scope of inference.